

Temujin Orkhon

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Education

University of California, Berkeley

PhD Candidate in Chemistry

Concentration: Quantum Chemistry, Computational Chemistry, Quantum Computing

Berkeley, CA

Expected June 2031

Massachusetts Institute of Technology

B.S. in Chemistry & Biology, B.S. in Physics, Minor in Computer Science

GPA: 5.0/5.0

Cambridge, MA

June 2026

Experience

Learning Matters Lab (Gómez-Bombarelli Group)

Undergraduate Researcher

Cambridge, MA

January 2023- June 2024

- Constructed multi-layered machine learning models, utilizing uncertainty as a learnable parameter, achieving greater generalizability and accuracy in predicting molecular property
- Developed software aimed at generating artificial datasets, implementing, and benchmarking existing molecular property machine learning models

Elkin Lab

Undergraduate Researcher

Cambridge, MA

January 2025- June 2025

- Designed novel algorithms to extract organic transformation templates, improving on literature template extractors through the usage of context-aware reaction centers
- Helped in the development of software that streamlines the analysis of total synthesis data to determine impactful multistep transformations for further experimental optimization

Van Voorhis Group

Undergraduate Researcher

Cambridge, MA

June 2024- June 2026

- Developed electronic structure methods employing quantum embedding to calculate the electronic properties of strongly correlated chemical systems.
- Simulated and analyzing defected semiconductor materials for applications in quantum computing, collaborating with outside universities

Publications & Presentations

- Orkhon T.**; Weatherly S.; Van Voorhis T. (2025), "Extracting band structures using Bootstrap Embedding
- Contributor** (2025), "QuEmb: A Toolbox for Bootstrap Embedding Calculations of Molecular and Periodic Systems", Journal of Physical Chemistry, doi.org/10.1021/acs.jpca.5c02983
Contributed by testing the periodic bootstrap embedding code, and helping with bug fixes
- Orkhon T.**; Rein J.; Elkin M. (2025), "Holistic reaction template generation using a context-driven approach" Presented at the Chem UROP Symposium, Cambridge, MA
- Orkhon T.**; Greenman K.; Green W.; Gómez-Bombarelli R. (2024) "Uncertainty improves multi-fidelity machine learning", Presented at the ChemE UROP Symposium, Cambridge, MA
- Greenman K.; **Orkhon T.**; Green W.; Gómez-Bombarelli R. (2023), "Multi-fidelity deep learning for data-efficient molecular property models from experimental and computational data", AIChE Annual Meeting

Teaching Experience

MIT Department of Chemistry

Chemistry Tutor

Cambridge, MA

February 2024- June 2026

- Provided one-on-one tutoring in general chemistry, organic chemistry, and physical chemistry to MIT students interested in chemistry.
- Created custom lesson plans, practice problems, and review materials tailored to the student's needs.

Talented Scholars Resource Room

Facilitator

Cambridge, MA

Sept. 2023- June 2024

- Facilitated student groups, providing comprehensive assistance with problem sets and hosting exam reviews, improving academic performance of students at MIT.
- Delivered personalized one-on-one guidance to MIT students in fundamental introductory subjects, ensuring an effective approach.

TomYo EdTech

College Counselor

Ulaanbaatar, Mongolia

August 2023- Sept. 2024

- Advocated to Mongolian students to study abroad, guiding high school students through the college application process
- Counseled prospective students on their higher education plans, offering valuable insights and guidance to align their goals with suitable academic pathways.

Honors & Awards

- Sigma Pi Sigma (Physics Honor Society)
- Academic achievement award (MIT Chemistry 2026)
- Miyagawa Prize winner (MIT Global Languages 2026)
- Best poster presentation in the ChemE UROP Symposium (2024)
- International Chemistry Olympiad Silver medalist (2021, 2022)
- International Mendeleev Chemistry Olympiad Silver medalist (2021)

Technical & Laboratory Skills

Languages: English, Mongolian, Japanese

Computational: Python, Pandas, Numpy, PyTorch, SciPy, FHI-aims, PySCF, QuEmb, QChem

Laboratory: Titration, Distillation, Thin-Layer Chromatography, Column Chromatography, UV-vis Spectroscopy, Fluorescence Spectroscopy, FTIR Spectroscopy

Interests/Hobbies: Photography, Football (Soccer), Basketball